

A Practical Guide to Autonomous vehicle

TOPIC -- AUTONOMOUS VEHICLE

-- 01 --

2010s In a 2011 online survey of 2,006 US and UK consumers, 49% said they would be comfortable using a "driverless car". A 2012 survey of 17,400 vehicle owners found 37% who initially said they would be interested in purchasing a "fully autonomous car". However, that figure dropped to 20% if told the technology would cost US\$3,000 more. In a 2012 survey of about 1,000 German drivers, 22% had a positive attitude, 10% were undecided, 44% were skeptical and 24% were hostile. A 2013 survey of 1,500 consumers across 10 countries found 57% "stated they would be likely to ride in a car controlled entirely by technology that does not require a human driver", with Brazil, India and China the most willing to trust automated technology. In a 2014 US telephone survey, over three-quarters of licensed drivers said they would consider buying a self-driving car, rising to 86% if car insurance were cheaper. 31.7% said they would not continue to drive once an automated car was available. In 2015, a survey of 5,000 people from 109 countries reported that average respondents found manual driving the most enjoyable. 22% did not want to pay more money for autonomy. Respondents were found to be most concerned about hacking/misuse, and were also concerned about legal issues and safety. Finally, respondents from more developed countries were less comfortable with their vehicle sharing data. The survey reported consumer interest in purchasing an AC, stating that 37% of surveyed current owners were either "definitely" or "probably" interested. In 2016, a survey of 1,603 people in Germany that controlled for age, gender, and education reported that men felt less anxiety and more enthusiasm, whereas women showed the opposite. The difference was pronounced between young men and women and decreased with age. In a 2016 US survey of 1,584 people, "66 percent of respondents said they think autonomous cars are probably smarter than the average human driver". People were worried about safety and hacking risk. Nevertheless, only 13% of the interviewees saw no advantages in this new kind of cars. In a 2017 survey of 4,135 US adults found that many Americans anticipated significant impacts from various automation technologies including the widespread adoption of automated vehicles. In 2019, a test of a method of surveying opinions about self-driving cars, termed the Autonomous Vehicle Acceptance Model (AVAM), found that users were less accepting of high autonomy levels and displayed significantly lower intention to use highly autonomous vehicles. Additionally,

partial autonomy (regardless of level) was perceived as requiring uniformly higher driver engagement (usage of hands, feet and eyes) than full autonomy.

-- 02 --

NIO Navigate on Pilot On 12 August 2021, a 31-year-old Chinese man was killed after his NIO ES8 crashed in a tunnel. NIO's self-driving feature was in beta and could not deal with static obstacles. The vehicle's manual clearly stated that the driver must take over near construction sites. Lawyers of the deceased's family questioned NIO's private access to the vehicle, which they argued did not guarantee the integrity of the data.

-- 03 --

Level 3 – Conditional automation As of April 2024, two car manufacturers have sold or leased Level 3 cars: Honda in Japan, and Mercedes in Germany, Nevada and California. Mercedes Drive Pilot has been available on the EQS and S-class sedan in Germany since 2022, and in California and Nevada since 2023. A subscription costs between €5,000 and €7,000 for three years in Germany and \$2,500 for one year in the United States. Drive Pilot can only be used when the vehicle is traveling under 40 miles per hour (64 km/h), there is a vehicle in front, readable line markings, during the day, clear weather, and on freeways mapped by Mercedes down to the centimeter (100,000 miles in California). As of April 2024, one Mercedes vehicle with this capability has been sold in California. As of 2026, the only level 3 cars in China are Deepal SL03 and Arcfox αS.

-- 04 --

According to SAE J3016, Some driving automation systems may indeed be autonomous if they perform all of their functions independently and self-sufficiently, but if they depend on communication and/or cooperation with outside entities, they should be considered cooperative rather than autonomous.

-- 05 --

Notification In accordance with the proposed SAE J3134 "ADS Marker Lamps" specification, the US has standardized the use of turquoise lights to inform other drivers that a vehicle is driving autonomously. It will be used in the 2026 Mercedes-Benz EQS and S-Class sedans with Drive Pilot, an SAE Level 3 driving system. As of 2023, the turquoise light had not been standardized in China or by the UNECE.

-- 06 --

In the 2020s In 2022, a survey found that 27% of the world's population would feel safe in a self-driving car. A July 2024 biannual YouGov survey of British adults found that 4% would feel "very safe" in a self-driving car while 37% would feel "very unsafe".

-- 07 --

Trolley problem The trolley problem is a thought experiment in ethics. Adapted for ACs, it considers an AC carrying one passenger confronting a pedestrian who steps in its way. The ADAS notionally has to choose between killing the pedestrian or swerving into a wall, killing the passenger. Possible frameworks include deontology (formal rules) and utilitarianism (harm reduction). One public opinion survey reported that harm reduction was preferred, except that passengers wanted the vehicle to prefer them, while pedestrians took the opposite view. Utilitarian regulations were unpopular. Additionally, cultural viewpoints exert substantial influence on shaping responses to these ethical quandaries. Another study found that cultural biases impact preferences in prioritizing the rescue of certain individuals over others in car accident scenarios.

-- 08 --

Driving systems Advanced driver-assistance systems (ADAS) automate specific driving features such as Forward Collision Warning (FCW), Automatic Emergency Braking (AEB), Lane Departure Warning (LDW), Lane Keeping Assistance (LKA) or Blind Spot Warning (BSW). An ADAS requires a human driver to handle tasks that the ADAS does not support. ADAS contrasts to an automated driving system (ADS), which would be classified by SAE J3016 as Level 3 or higher.

-- 09 --

Trust Consumers will avoid ACs unless they trust them as safe. Robotaxis operating in San Francisco received pushback over perceived safety risks. Automatic elevators were invented in 1900, but did not become common until operator strikes and trust was built with advertising and features such as an emergency stop button. However, with repeated use of autonomous driving functions, drivers' behavior and trust in autonomous vehicles gradually improved and both entered a more stable state. At the same time, this also improved the performance and reliability of the vehicle in complex conditions, thereby increasing public trust.